

Project Resources

1. The name of your project and the names of your team members.
2. A short description of your project.
3. A list of the technical features you plan to implement (this can be tentative at this stage).
4. An overview of the creative aspects of your project.
5. Some other comments (in case you have alternative ideas you are debating).

We encourage you to be creative, innovative, and utilize the technical knowledge.

Extra credit criteria: <https://canvas.ucsd.edu/courses/74034/assignments/1117765>

Final project pdf: <https://canvas.ucsd.edu/courses/74034/assignments/1117771>

Blog #1

Project Name: CloserXR - Sales Negotiator

Team Members

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Description

- Following Theme 1 (Conversation-Aware Bodily Gestures), CloserXR is an interactive MR experience where you get pitched a sales idea by a super charismatic, “Wolf of Wall Street”-style AI. We want to connect the Gemini LLM to a 3D avatar, emphasizing its body language to sell a fake product. The main idea we are trying to explore is how body language can make an AI feel more persuasive and real. We will be using Mixamo for all the animation clips and setting up an animator state machine to trigger the right movements based on what the AI says. We plan to keep the video passthrough mode that we enabled in Project 3, as well as the spatial anchor and modeling system we built to outline the key meshes like the floor and walls. The movement and body language will be driven by the conversational flow through the Gemini RESTful API, and we will parse the LLM’s text output for specific keywords or emotions. For example, if the user uses the microphone to push back on the price, the agent will shift into a defensive stance. Vice versa, if he is close to making the sale, the agent will do an intense pointing gesture to celebrate. Overall, we plan to create an energetic and enthusiastic AI sales agent with a heavy focus on body language.

Technical features

- **Passthrough and Spatial Anchors:** The agent will be able to co-exist in the real world. It will have the same interaction capabilities and environment mapping with what we did in Project 3 so the salesman can pace around in the real room and objects.
- **Gemini Restful API:** We will connect the Quest 2 to Google AI Studio so Gemini can handle the live sales dialogue and then write a script to read that text so the system knows when the user is pushing back or agreeing to the sale.
- **Mixamo Animator State:** We will take the animation clips from Mixamo and build a state machine that creates the dramatic pointing, pacing or defensive stances dynamically based on what Gemini says.
- **Speech Input:** Since the user would need to talk to the agent, we would set up some kind of speech tool to capture the user’s audio so we can send it to the LLM and create dialogue.

Overview of the creative aspects

- **Dialogue:** We want the dialogue to be as dramatic and enthusiastic as possible. We want the interaction to feel like a high-stakes negotiation game and the user can reject sales and say “This is too expensive” then the avatar will switch its stance and try to win the user back over.
- **Confidence.** The main goal is to show that body language can be used to show emotion and to make it feel real. By using the “Wolf of Wall Street” style, the agent can perform actions that are confident like using sharp hand pointing or intense pacing to make the user feel it’s a real pitch.
- **Spatial Proximity:** We want the agent to be able to interact with the physical space. This mean that when the agent is confident and trying to close a deal, it will move closer to the user to indicate that they should take the deal and to pressure them. So the agent would be able to do some pacing around while interacting with the physical space around it.

Comments

- **Model Choice:** We will start off using Gemini as our main foundational model, but will also consider other LLM options such as Grok and OpenAI.
- **Alternative versions:**
 - A car salesman is trying to sell the user a luxury car.
 - A startup founder pitching the user an investment.
 - A real estate agent selling a high-risk property deal.
 - A financial guru trying to sell an expensive course.
 - Best pick: AI sales automation tool or luxury investment opportunity, because it feels modern and lets the agent be dramatic without needing many physical props.